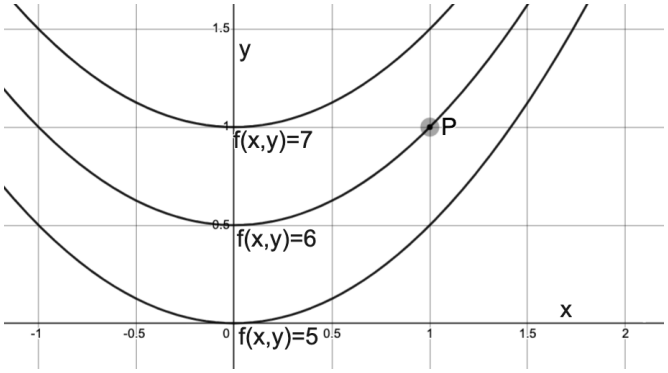
Math 119 Midterm **Winter 2024**

Q1. Short answer **Write your answer in the box. A correct answer in the box will earn full marks, regardless of any work shown. An incorrect answer may earn part marks for work shown.**

1. **[2 marks]** Let $f(x,y) = 2x^2 + y^2$ and let $\vec{p} = (1,1)$. There are two unit vectors $\vec{u}$ such that $D_{\vec{u}}f(\vec{p}) = 0$: find both such $\vec{u}$.

2. **[2 marks]** Let $f(x,y) = e^{3x}(y^2 - x^2)$. Using the tangent plane of the surface $z = f(x,y)$ at the point $(x,y,z) = (0,2,f(0,2))$, estimate $f(0.1, 1.9)$.

3. **[2 marks]** Shown below are 3 level curves of a surface $z = f(x,y)$. A point P is indicated. Determine if $f_x(P)$ is positive or negative, and determine if $f_y(P)$ is positive or negative.

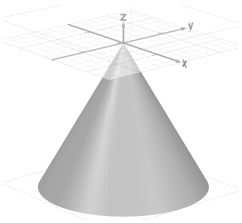


4. [**2 marks**] There are real numbers a, b and functions $g(y), h(y)$ such that

$$\int_{-2}^{0}\int_{x^2}^{4} f(x,y)\,dy\,dx = \int_{a}^{b}\int_{g(y)}^{h(y)} f(x,y)\,dx\,dy$$

regardless of what $f(x,y)$ is. Determine $a, b, g(y), h(y)$.

5. [**2 marks**] The graph below is a plot of $\nabla f(x,y)$ on the region $0 \le x, y \le 4$ for some function $f(x,y)$. Sketch the contour line (level curve) of $f(x,y)$ that goes through the point $(x,y) = (2,1)$. [**You may draw your answer directly on the graph above, or draw your own graph below - if there are multiple graphs, clearly indicate which one is your answer.**]

6. 

[**4 marks**] Let $\mathcal{D}$ be the cone in $\mathbb{R}^3$ defined by

$$-1 \leq z \leq -\sqrt{x^2+y^2}$$

- an image of $\mathcal{D}$ is shown to the left. Write $\iiint_{\mathcal{D}} x\,dV$ as an integral in each of cylindrical and spherical coordinates. [**Note: You do not need to evaluate these integrals. Also, you do not need to 'simplify' any trig expressions you get - having** $\sin(\frac{\pi}{2}-x)$ **in your answer would be fine, for example.**]

Q2. Let $f(x, y) = x^2 - 2x + 2y^2$.

(a) [**3 marks**] Find and classify the critical points of $f(x, y)$ using the Second Derivative Test.

(b) [**4 marks**] Subject to the constraint $x^2 + y^2 = 4$, use the Method of Lagrange Multipliers to determine the points (x, y) for which the maximum and minimum values of $f(x, y)$ occur.

(c) [**2 marks**] On the region $x^2 + y^2 \leq 4$, what are the maximum and minimum values of $f(x, y)$?

Q3. Consider an insect moving around on a parametric curve $\vec{\alpha}(t)$ defined by

$$\vec{\alpha}(t) = (x(t), y(t)) = (1 + \sin t, 1 + \sin^2 t)$$

where we consider over the time interval $0 \leq t \leq \pi$.

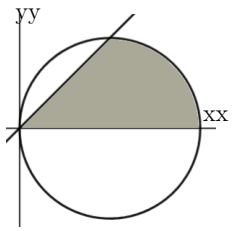
1. [**4 marks**] Provide a detailed sketch of the path taken by the insect. Your answer should include a description of a curve that the insect is always on - this will be a relationship between x and y that doesn't involve any t's.

2. [**4 marks**] Let $T(x, y)$ be the the air temperature of the room that the insect is in, as a function of x, y. Thus, the air temperature of the location of the insect at time t, which we call $g(t)$, is given by $g(t) = T(x(t), y(t))$ where $x(t), y(t)$ are the functions from above. Express $g'\left(\frac{\pi}{6}\right)$ in terms of the partial derivatives of T.

Q4. [6 marks] You only need to do one of (a) and (b). To earn full marks, you must (correctly) do (b). If you do (a), the maximum marks you will get is 4. If you attempt both, your best one will count.

(a) Let $\mathcal{A}$ be the region in $\mathbb{R}^2$ bounded by the four lines $y = x$, $y = x+2$, $y = 2x-8$, and $y = 2x - 4$. Invent a change of variables, with new coordinates u, v, that transforms $\mathcal{A}$ into a rectangular domain, and evaluate $\iint_{\mathcal{A}} (x - y)\, dx\, dy$ using the Change of Variables Theorem.

(b) Let $\mathcal{B}$ be the region in $\mathbb{R}^2$ bounded by the four curves $y = \dfrac{2}{x}$, $y = \dfrac{5}{x}$, $y = \dfrac{4}{x^2}$, and $y = \dfrac{7}{x^2}$. Invent a change of variables, with new coordinates u, v, that transforms $\mathcal{B}$ into a rectangular domain, and evaluate $\iint_{\mathcal{B}} xy\, dx\, dy$ using the Change of Variables Theorem.

Q5. 

[6 marks] Consider the region $\mathcal{D}$ defined by the inequalities $(x-1)^2 + y^2 \leq 1$, $x \geq y$, and $y \geq 0$ - an image of $\mathcal{D}$ is shown to the left - it is the shaded area. By doing a double integral using polar coordinates, determine the area of $\mathcal{D}$. [**Note**:An identity you may find useful is $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$. It is possible to obtain the correct answer without doing an integral - doing this will earn 0 marks.]

Q6. [**4 marks**] Consider the surface $z = f(x, y) = 1 - x^2 - y^2$. For each plane which is tangent to (a point on) the surface, and which contains the points $(1, 0, 2)$ and $(0, 2, 2)$, determine an equation of that plane.

Q7. [**3 marks**] The plane $x + y + z = 1$ divides the solid sphere

$$x^2 + y^2 + z^2 \leq 1$$

into two regions with different volumes. Let $\mathcal{D}$ be the region with the smaller volume. By using either cylindrical or spherical coordinates, evaluate $\iiint_{\mathcal{D}} 1 \, dV$ [**Note**: This question is meant to be very challenging and is only worth 3 marks - we recommend being happy with everything else before attempting it.]